

# A Literature-Implied Upper Bound for $H'_\alpha(\ell_1^4)$

Run `fa.banach_001`, lane 9

2026-06-19

## Status

This is a lightweight literature-implied partial-subcase note for Mercourakis–Vassiliadis, arXiv:1902.05593. It records an implication of a later Banach–Mazur distance theorem of Kobos–Varivoda, arXiv:2305.06427. It is not an original full solution of the source paper’s open-problem list.

## Source Question

Mercourakis and Vassiliadis conclude with open problems. Item 1(a) asks for better upper and lower bounds for

$$H'_\alpha(\ell_p^n) \quad \text{and} \quad H_\alpha(\ell_p^n)$$

for  $n \geq 4$  and  $1 \leq p < 2$ . The subcase recorded here is the strict antipodal number of  $\ell_1^4$ .

## Supporting Literature

Kobos and Varivoda prove in Theorem 3.1 of [2] that the Banach–Mazur distance between the four-dimensional cube  $C_4$  and its polar cross-polytope  $C_4^*$  is exactly 2. In Banach-space notation,

$$d_{\text{BM}}(\ell_\infty^4, \ell_1^4) = 2.$$

## Identification

We claim that the supporting theorem implies

$$H'_\alpha(\ell_1^4) < 16,$$

and hence, since  $H'_\alpha$  is integer-valued,

$$H'_\alpha(\ell_1^4) \leq 15.$$

Indeed, suppose toward a contradiction that  $H'_\alpha(\ell_1^4) = 16 = 2^4$ . Then there is a strict bounded separated antipodal set  $S \subset S_{\ell_1^4}$  with 16 points and separation constant  $d > 1$  in the sense of the source paper.

The renorming theorem quoted in the introduction of [1] says that such a set becomes equilateral under an equivalent norm  $||| \cdot |||$ , and the Banach–Mazur distance from the original norm is at most  $2/d$ . Thus

$$d_{\text{BM}}((\mathbb{R}^4, \|\cdot\|_1), (\mathbb{R}^4, |||\cdot|||)) \leq \frac{2}{d} < 2.$$

Since an equilateral set in a four-dimensional normed space has at most  $2^4$  points, with equality only for  $\ell_\infty^4$ , the renormed space is linearly isometric to  $\ell_\infty^4$ . Consequently

$$d_{\text{BM}}(\ell_1^4, \ell_\infty^4) < 2,$$

contradicting Kobos–Varivoda’s equality  $d_{\text{BM}}(\ell_1^4, \ell_\infty^4) = 2$ .

## Scope

This only improves the upper bound in the single endpoint subcase  $(p, n) = (1, 4)$  for the strict antipodal Hadwiger number. It does not determine the exact value of  $H'_\alpha(\ell_1^4)$ , does not decide  $H_\alpha(\ell_1^4)$ , and does not address the remaining  $1 < p < 2$  cases. The relation appears to be an agent-identified implication rather than a claim explicitly made by Kobos–Varivoda.

## Search Notes

Cheap run-index searches for 1902.05593, antipodal Hadwiger, Hadwiger numbers, and finite-dimensional Banach keywords found no existing packet for this source paper. Targeted web searches for antipodal Hadwiger phrases found the source paper, while searches for Banach–Mazur distance between  $\ell_1^4$  and  $\ell_\infty^4$  found Kobos–Varivoda’s later paper. No direct later paper stating this exact  $H'_\alpha(\ell_1^4)$  corollary was found in this bounded search.

## References

- [1] S. K. Mercourakis and G. Vassiliadis, *Antipodal Hadwiger numbers of finite-dimensional Banach spaces*, arXiv:1902.05593, 2019.
- [2] T. Kobos and M. Varivoda, *On the Banach–Mazur Distance in Small Dimensions*, arXiv:2305.06427, 2023.